

Kunal Pai

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EDUCATION

Ph.D., Computer Science , University of California, Los Angeles	Enrolled: September 2026
M.S., Computer Science , University of California, Davis (GPA: 4.0/4.0)	June 2026
B.S., Computer Science & Engineering , University of California, Davis (GPA: 3.8/4.0), with Honors	June 2023

RELEVANT SKILLS

Languages: Python, C++, C, Rust, JavaScript, TypeScript, Java, Bash

ML/AI: LLMs, Multi-Agent Systems, AI Agent Security, Prompt Engineering, PyTorch, TensorFlow, Hugging Face, Ollama, scikit-learn, Model Calibration, Evolutionary Algorithms

Systems & Compilers: gem5, Computer Architecture Simulation, RISC-V, LLVM, Clang, C-to-Rust Transpilation, Hardware Profiling (perf)

Web/Data: React, Next.js, Django, Flask, MongoDB, pandas, NumPy, Matplotlib, REST APIs

Tools: Git, Docker, Unix/Linux, Jupyter, LaTeX, Slurm / HPC, GitHub Actions

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Student Researcher, DavSec Lab @ UC Davis

April 2025 – June 2026

Research Project

Python, LLMs, C++, Rust, Compiler Analysis

- Built automated C-to-Rust transpilation pipeline using LLMs with 5 prompt variations, benchmarking state-of-the-art models across 746 programs and achieving 70.2% functional accuracy
- Engineered a multi-variant performance-aware transpilation framework that explores alternative Rust implementations in an evolutionary agent setting, yielding up to 3× runtime reduction across benchmarks
- Developed cross-layer analysis framework combining compiler metrics with hardware performance counters to expose semantic and optimization gaps in LLM-translated, syntax-directed and manually-written Rust translations

Graduate Student Researcher, DECAL Lab @ UC Davis

September 2022 – December 2024

Research Project

Python, LLMs, Machine Learning, Code Analysis

- Developed a 4,500-sample dataset for pairwise code-documentation alignment from 200 open-source Python projects, enabling future research in software maintenance
- Engineered a pipeline for measuring calibration and correctness of large language models for code repair, using Defects4J
- Quantified precision and recall gains (ROUGE, METEOR) from semantic augmentation in LLM prompt pipelines for code summarization

Teaching Assistant, University of California, Davis

September 2023 – December 2023

- Assisted 180 students in a senior-level Probability & Statistical Modeling class.

PUBLICATIONS (SELECTED)

VATS: Exploiting Implicit Authority in Error-Path Injection via Systematic Mutation [SPOTLIGHT], Patel, H. & Pai, K., *Second Workshop on Agents in the Wild: Safety, Security, and Beyond, ICML 2026*

Toward Reproducible and Standardized Computer Architecture Simulation with gem5, Pai, K., Patel, H., Le, E., et. al., *IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2026*

NAAMSE: Framework for Evolutionary Security Evaluation of Agents, Pai, K., Shah, P. & Patel, H., *ICLR 2026 Agents in the Wild: Safety, Security, and Beyond Workshop*

Implications of Full-System Modeling for Superconducting Architectures, Pai, K., Samani, M., Nand, A. & Lowe-Power, J., *Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops '25)*

CoDocBench: A Dataset for Code-Documentation Alignment in Software Maintenance, Pai, K., Devanbu, P. & Ahmed, T., *International Conference on Mining Software Repositories (MSR) 2025: Data and Tool Showcase Track*

Calibration and Correctness of Language Models for Code, Spiess, C., Gros, D., Pai, K.S., et. al., *International Conference on Software Engineering (ICSE) 2025*

Automatic semantic augmentation of language model prompts (for code summarization), Ahmed, T., Pai, K.S., Devanbu, P. & Barr, E.T., *International Conference on Software Engineering (ICSE) 2024*

PROJECT EXPERIENCE

NAAMSE: Neural Adversarial Agent Mutation-based Security Evaluator <i>Research Project</i>	Nov 2025 – Present <i>Python, LLMs, Evolutionary Algorithms</i>
• Won 2nd place (Agent Safety) at UC Berkeley RDI AgentBeats Competition; framework infrastructure was subsequently forked by Mozilla's Odin team • Designed and implemented a clustering engine to identify semantic attack vectors in LLM-generated adversarial prompts • Engineered an attack pipeline that iteratively generates, evaluates, and refines adversarial prompts against target models • Benchmarked multiple frontier LLMs within the framework to validate attack effectiveness and refine scoring metrics	
HASHIRU: Hierarchical, Resource-Aware Multi-Agent Framework <i>Research Project</i>	March 2025 – June 2025 <i>Python, LLMs, Multi-Agent Systems</i>
• Designed and deployed a multi-agent architecture enabling dynamic, LLM-driven collaboration across diverse tasks • Implemented task decomposition with intelligent agent delegation based on resource cost models and task specialization • Engineered autonomous generation of tools and APIs for task execution • Developed a robust evaluation framework for agent performance across complex, multi-step tasks	
SuperNOVA: Superconducting Graph Accelerator in gem5 <i>Research Project, DArchR Lab @ UC Davis</i>	January 2024 – March 2026 <i>C++, Python, gem5, Hardware Simulation</i>
• Modeled a superconducting graph accelerator and interconnect in gem5, achieving up to 24× speedup and 73× energy efficiency • Integrated cryogenic/superconducting cores, caches, and interconnects to evaluate bottlenecks for realistic workloads • Mentored undergraduates on modeling, benchmarking, and research writing; one co-authored a ModSim 2024 poster	
gem5 Vision <i>Resource Discovery Framework</i>	January 2023 – June 2023 <i>Next.js, Python, MongoDB, JSON Schema</i>
• Accelerated resource discovery 20× for 1,200+ gem5 artifacts by optimizing search and categorization logic • Implemented categorization and semantic versioning across 20+ resource types to streamline retrieval • Integrated local and remote JSON schemas with MongoDB, improving accessibility for 500+ users	

AWARDS AND HONORS

Finalist (Agentic Financial & High-Stakes AI Advice) , Cambridge C:\>DIR Global 'Agentic Regulator' Hackathon	2026
2nd Place (Agent Safety) , UC Berkeley RDI AgentBeats Competition	2026
Dean's List , UC Davis College of Engineering	Fall 2019, Fall 2020, Winter 2022, Spring 2022
Provost Award , UC Davis	2019–2023

SERVICE

Program Committee
• MSR, Data and Tool Showcase Track (2027) • MSR, Data and Tool Showcase Track (2026)
Artifact Evaluation Committee
• ISSTA (2026) • ASE (2026) • ISPASS (2026)
Reviewer
• NeurIPS, CODEC-FM Workshop (2026) • NeurIPS, SocialAgent Workshop (2026) • NeurIPS, Agents in the Wild Workshop (2026) • ICML, Agents in the Wild Workshop (2026)